# 深圳市宏翔瑞科技有限公司

# 产品承认书

客户名称
Customer:

产品描述
DESCRIBE: 7" TFT

产品型号
PRODUCTS:

液晶玻璃
FOG OF LCD: IPS

质量等级
class of commodity: A 级 （ 带点率≦5% ）

质量保证
quality assurance: 保证一年

| | |
|---|---|
| **供 应 商** | |
| **地 址** | |
| **电 话** | |
| **传 真** | |
| **日 期** | |

## REVISION STATUS

| Version | Revise Date | Page | Content | Modified by |
|---|---|---|---|---|
| V1.0 | 2020.03.27 | - | First Issued. | LUANJUN |

# TABLE OF CONTENTS


| NO. | CONTENTS | PAGE |
|---|---|---|
| | REVISION STATUS | 2 |
| | TABLE OF CONTENTS | 3 |
| 1. | GENERAL DESCRIPTION | 4 |
| 2. | MECHANICAL SPECIFICATION | 5 |
| 3. | PIN DESCRIPTION | 6 |
| 4. | ELECTRICAL CHARACTERISTICS | 8 |
| 5. | TIMING CHARACTERISTICS OF INPUT SIGNALS | 11 |
| 6. | TIMING SEQUENCE(TIMING CHART) | 13 |
| 7. | OPTICAL CHARACTERISTICS | 16 |
| 8. | RELIABILITY TEST ITEMS | 19 |
| 9. | GENERAL PRECAUTION | 20 |
| 10. | PACKAGE DRAWING | 21 |

# 1. GENERAL DESCRIPTION

## 1.1 DESCRIPTION

070WSI30-2721 is a color active matrix thin film transistor (TFT) TN liquid crystal display (LCD) that uses amorphous silicon TFT as a switching device. It is composed of a TFT LCD panel, Driver IC ,FPC and Backlight.

## 1.2 FEATURES:

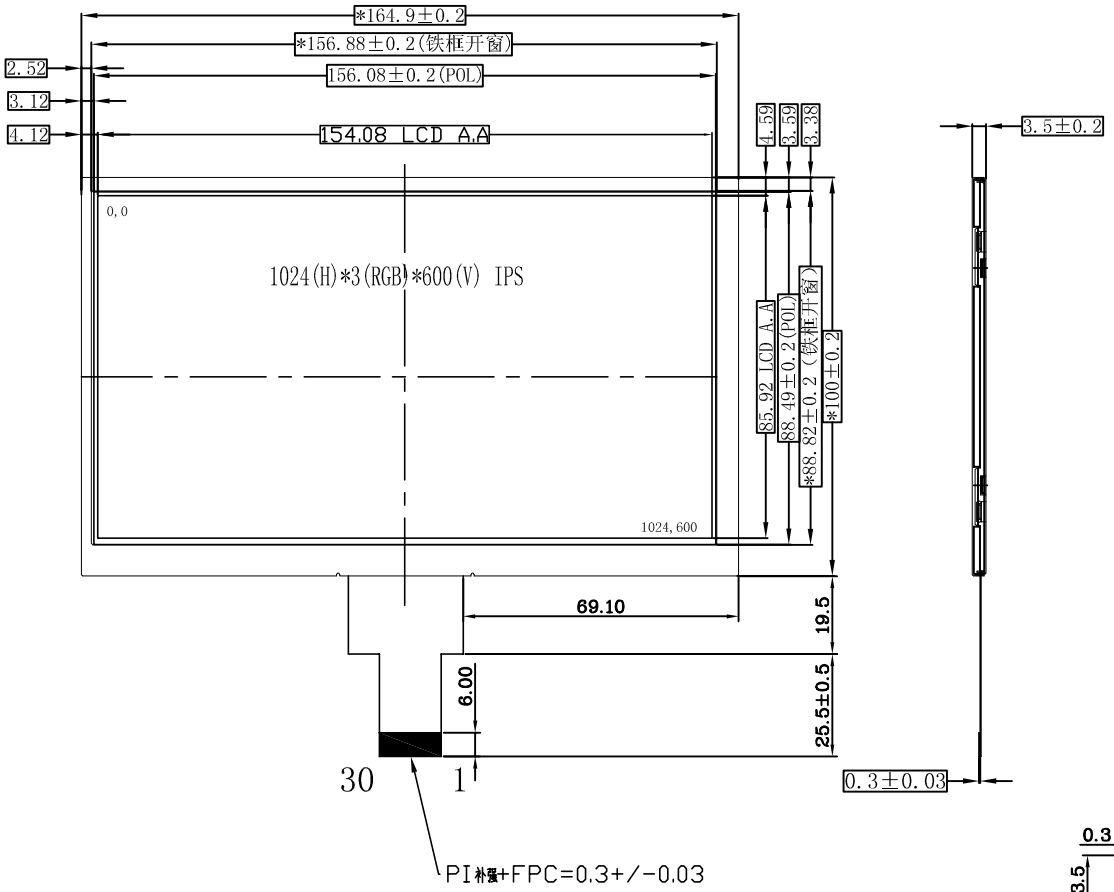
| No. | Item | Specification | Unit |
|---|---|---|---|
| 1 | Panel Size | 7” | inch |
| 2 | Number of Pixels | 1024×RGB (3) ×600 | pixels |
| 3 | Active Area | 154.21(H)x 85.92(V) | mm |
| 4 | Pixel Pitch | 0.1506(H)×0.1432(V) | mm |
| 5 | OutlineDimension | 165(W)×100(H)×3.5(D) | mm |
| 6 | Number of Colors | 16.7M | - |
| 7 | Display Mode | Normally black | - |
| 8 | Viewing Direction | 12 | - |
| 9 | Display Format | RGB vertical stripe | - |
| 10 | Luminance(cd/m^2) | 500 (TYP.) | net |
| 11 | Contrast Ratio | 600(TYP.) | |
| 12 | Surface Treatment | Anti-Glare | - |
| 13 | Interface | MIPI | - |
| 14 | Backlight | White LED | - |
| 15 | OperationTemperature | -20~70 | ℃ |
| 16 | StorageTemperature | -30~80 | ℃ |
| 17 | Polarizer Type | AG: Anti Glare (2H,3H)<br>HC: Hard Coating | |

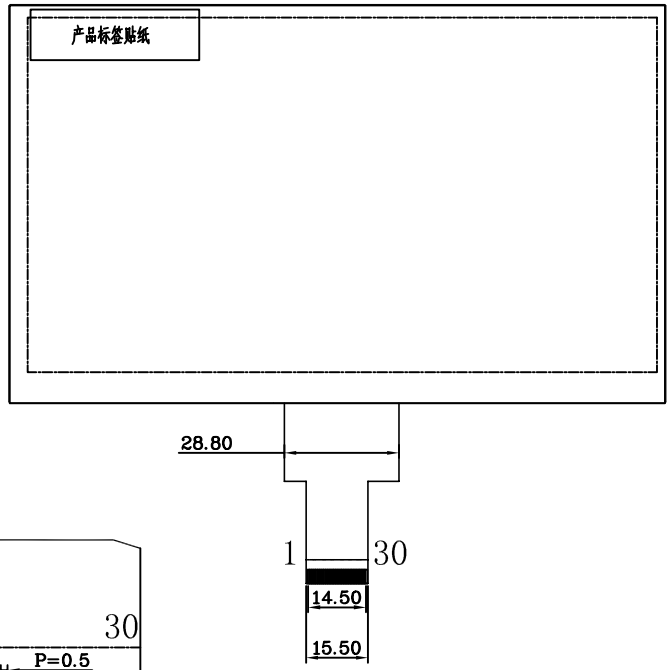
# Outline Drawing

| 客户确认签名 Customer approved | | 日期 Date | |
|---|---|---|---|

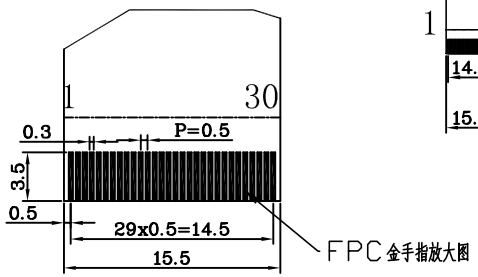
RoHS

| VER | REVISION RECORD | Date | Name |
|---|---|---|---|
| A | NEW RELEASE | 2021/10/09 | ZHAO |
| B | | | |
| C | | | |



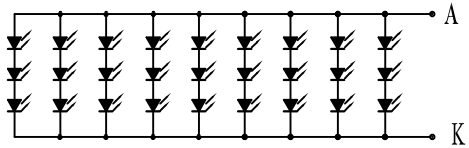




| NO. | PING NAME |
|---|---|
| 1 | VLED+ |
| 2 | VLED+ |
| 3 | LCD_VGH |
| 4 | LCD_VGL |
| 5 | UP/DN |
| 6 | SH/LR |
| 7-8 | VLED- |
| 9 | AVDD |
| 10 | GND |
| 11 | MP-TDP3 |
| 12 | MP-TDN3 |
| 13 | GND |
| 14 | MP-TDP2 |
| 15 | MP-TDN2 |
| 16 | GND |
| 17 | MP-CLKP |
| 18 | MP-CLKN |
| 19 | GND |
| 20 | MP-TDP1 |
| 21 | MP-TDN1 |
| 22 | GND |
| 23 | MP-TDP0 |
| 24 | MP-TDN0 |
| 25 | GND |
| 26 | STBYB |
| 27 | REST |
| 28-29 | VDD |
| 30 | VCOM |
| | |
| | |




NOTE

1.Lcd Display Type: 7.0'TFT,1024*600,IPS
2.Polarzier Mode:Transmissive,normally black
3.Viewing Direction: ALL
4.Logic Supply Voltage: 3.0V
5.Control IC: EK79007A+EK73217A
6.Backlignt: White(3*9 Led),ILed=180mA,Vled=9.6-10.0V
7.Backlight brightness and uniformity: 450cd/m2,>=80%
8.Operating Temp: -20 TO 70
9.Storage Temp：-30 TO 80℃
10.Connector: FPC
11.Ulteriorly Tolerance:±0.2MM



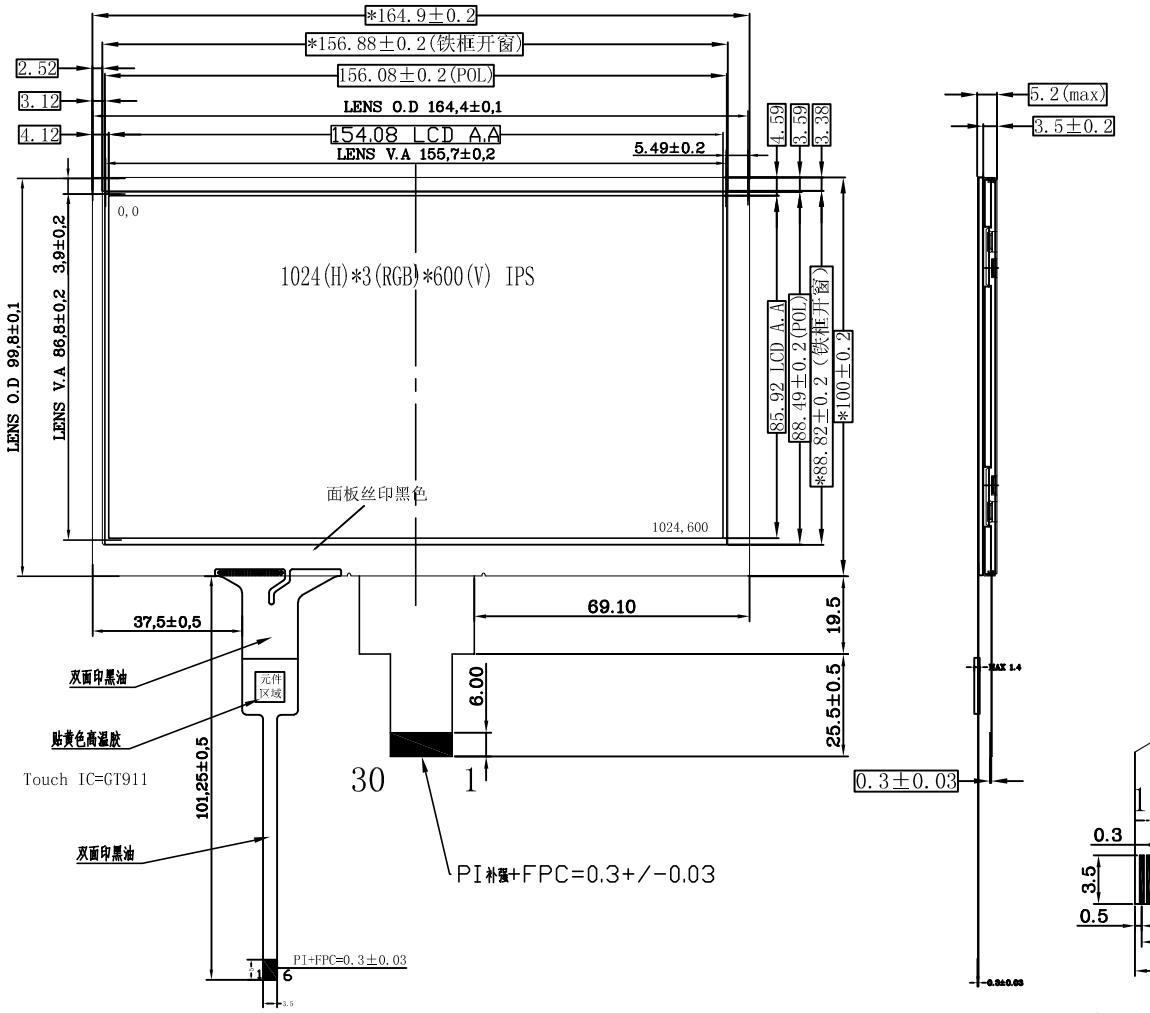

LED 电路图

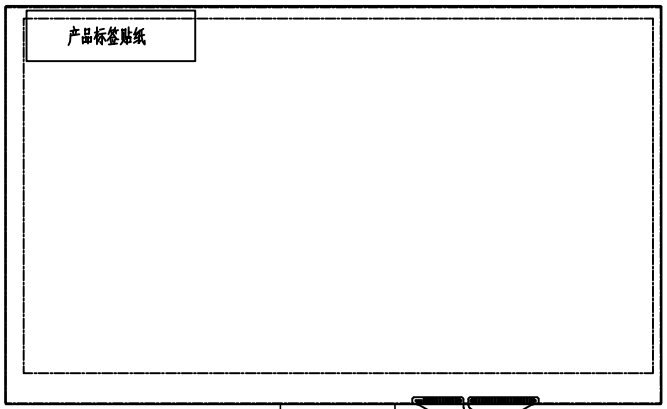
| 深圳市宏翔瑞科技有限公司 | | | | LCM TYPE 7.0吋 TFT |
|---|---|---|---|---|
| Dwg: | | CUSTOMER | | Material No. |
| Check: | | MODULE | | SCALE: 1:1 |
| Dim: MM | | Visual Angle | | SHEET1 OF 1 |

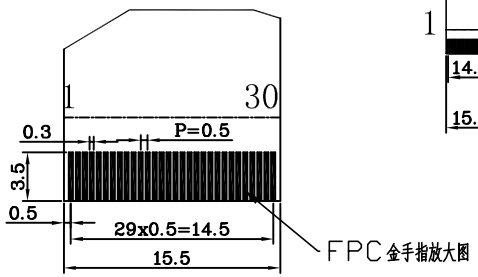
# Outline Drawing

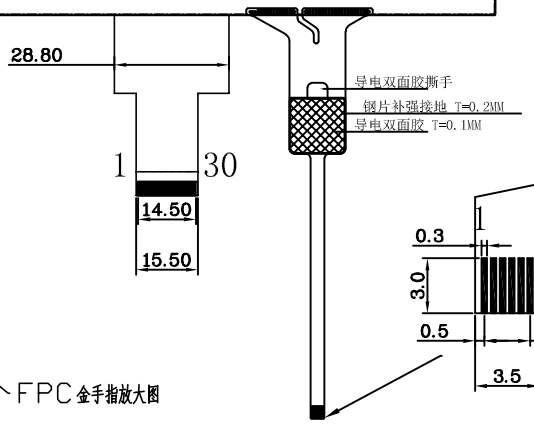
| 客户确认签名 Customer approved | | 日期 Date | |
|---|---|---|---|

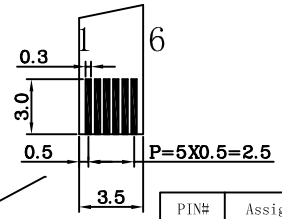
| VER | REVISION RECORD | Date | Name |
|---|---|---|---|
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| B | | | |
| C | | | |



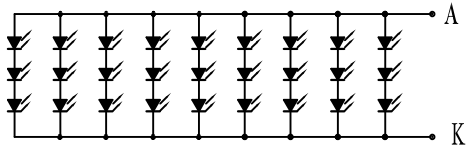




| NO. | PING NAME |
|---|---|
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| 2 | VLED+ |
| 3 | LCD_VGH |
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| 12 | MP-TDN3 |
| 13 | GND |
| 14 | MP-TDP2 |
| 15 | MP-TDN2 |
| 16 | GND |
| 17 | MP-CLKP |
| 18 | MP-CLKN |
| 19 | GND |
| 20 | MP-TDP1 |
| 21 | MP-TDN1 |
| 22 | GND |
| 23 | MP-TDP0 |
| 24 | MP-TDN0 |
| 25 | GND |
| 26 | STBYB |
| 27 | REST |
| 28-29 | VDD |
| 30 | VCOM |
| | |
| | |

| PIN# | Assigrment |
|---|---|
| 1 | Shutdown |
| 2 | VCC |
| 3 | GND |
| 4 | INT |
| 5 | SDA |
| 6 | SCL |

NOTE

1.Lcd Display Type: 7.0'TFT,1024*600,IPS
2.Polarzier Mode:Transmissive,normally black
3.Viewing Direction: ALL
4.Logic Supply Voltage: 3.0V
5.Control IC: EK79007A+EK73217A
6.Backlignt: White(3*9 Led),ILed=180mA,Vled=9.6-10.0V
7.Backlight brightness and uniformity: 350cd/m2,>=80%
8.Operating Temp: -20 TO 70
9.Storage Temp：-30 TO 80℃
10.Connector: FPC
11.Ulteriorly Tolerance:±0.2MM




LED 电路图

| 深圳市宏翔瑞科技有限公司 | | | | LCM TYPE 7.0 吋 TFT+CTP |
|---|---|---|---|---|
| Dwg: | | CUSTOMER | | Material No. |
| Check: | | MODULE | | SCALE: 1:1 |
| Dim: MM | | Visual Angle | | SHEET1 OF 1 |

## 3. PIN DESCRIPTION

| Pin No. | Symbol | Function | Remark |
|---|---|---|---|
| 1-2 | LED+ | LED Anode | |
| 3 | LVD-VGH | Gate ON Voltage | |
| 4 | LCD-VGL | Gate OFF Voltage | |
| 5 | UPDN | Gate Up or Down scan control | |
| 6 | SHLR | Source Right or Left sequence | |
| 7-8 | LED- | LED Cathode | |
| 9 | AVDD | Power for Analog Circuit | |
| 10 | GND | Ground | |
| 11 | MIPI-TDP3 | MIPI Data lane3 input | |
| 12 | MIPI-TDN3 | | |
| 13 | GND | Ground | |
| 14 | MIPI-TDP2 | MIPI Data lane2 input | |
| 15 | MIPI-TDN2 | | |
| 16 | GND | Ground | |
| 17 | MIPI-TCP | MIPI CLK input | |
| 18 | MIPI-TCN | | |
| 19 | GND | Ground | |
| 20 | MIPI-TDP1 | MIPI Data lane1 input | |
| 21 | MIPI-TDN1 | | |
| 22 | GND | Ground | |
| 23 | MIPI-TDP0 | MIPI Data lane0 input | |
| 24 | MIPI-TDN0 | | |
| 25 | GND | Ground | |
| 26 | STBYB | Standby mode | |
| 27 | LRSTB | Global reset pin | |
| 28-29 | VDD | Power Supply | |
| 30 | VCOM | Common Voltage | |

# 4. ELECTRICAL CHARACTERISTICS

## 4.1 ABSOLUTE MAXIMUM RATINGS

| Item | Symbol | Values | | Unit | Remark |
|---|---|---|---|---|---|
| | | Min. | Max. | | |
| Digital Supply Voltage | VDD | -0.3 | 5 | V | |
| Analog Supply Voltage | AVDD | -0.5 | 15 | V | |
| Gate On Voltage | VGH | -0.3 | 40 | V | |
| Gate Off Voltage | VGL | -20 | 0.3 | V | |
| Gate On-Gate Off Voltage | VGH-VGL | 12 | 40 | V | |

## 4.2 TFT LCD MODULE

### 4.2.1 Operating Conditions

| Parameter | Symbol | Min. | Typ. | Max. | Unit | Conditions |
|---|---|---|---|---|---|---|
| Digital Supply Voltage | VDD | 3.0 | 3.3 | 3.6 | V | |
| TFT Gate on voltage | VGH | 17 | 18 | 19 | V | |
| TFT Gate off voltage | VGL | -7 | -6 | -5 | V | |
| TFT Common electrode voltage | VCOM | 3.0 | 3.6 | 4.4 | V | |
| Analog power supply voltage | AVDD | 9.4 | 10 .4 | 10.6 | V | |

### 4.2.2 Current Consumption

| Item | Symbol | Condition | Values | | | Unit | Remark |
|---|---|---|---|---|---|---|---|
| | | | Min. | Typ. | Max. | | |
| Gate on power Current | IVGH | VGH =18 V | - | 0.5 | 1 | mA | |
| Gate off power current | IVGL | VGL= -6V | - | 1 | 2 | mA | |
| Digital power current | IVDD | VDD = 3.3V | - | 30 | 45 | mA | |
| Analog power current | IAVDD | AVDD =10.4V | - | 35 | 45 | mA | |

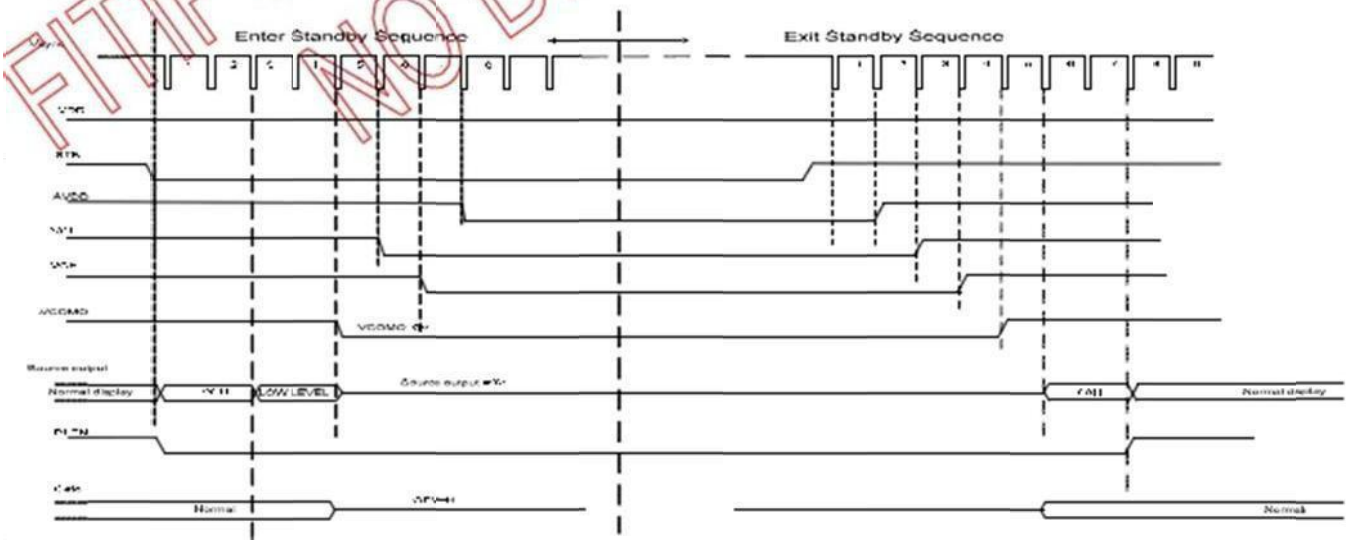
## 4.3 POWER ON/OFF SEQUENCE

In order to prevent IC from power on reset fail, the rising time (TPOR) of the digital power supply VDD should be maintained within the given specifications. Refer to " AC Characteristics" for more detail on timing.

### 4.3.1 POWER-ON/OFF TIMING SEQUENCE




Power On/Off timing chart




Enter and Exit Standby Mode timing chart

Note: Low level=3Fh,when NBW=L(Normally white)

Low level=00h,when NBW=H(Normally black)

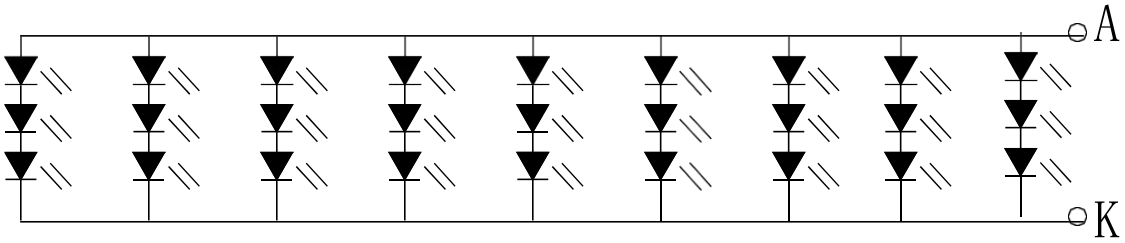
FITIPOWER CONFIDENTIAL
NO DISCLOSURE

## 4.4 BACK LIGHT UNIT

Ta=25℃

| Item | Symbol | Min. | Typ. | Max. | Unit | Remark |
|---|---|---|---|---|---|---|
| LED current | ILED | - | 180 | - | mA | 27LEDS |
| Forward voltage | VF | 8.4 | 9.0 | 9.6 | V | IF=180mA 27LEDS |
| Reverse current | IR | - | - | 50 | μA | VR=3V,1LED |
| Power dissipation | Pd | 1470 | | | mW | 27LEDS |
| Peak forward current | IFP | 20 | | | mA | 1LED |
| Reverse Voltage | VR | 3 | | | V | 1LED |

### 4.4.1 Internal Circuit Diagram




3X9=27LED 9.0V

# 5. TIMING CHARACTERISTICS OF INPUT SIGNALS

## 5.1 PARALLEL RGB INPUT TIMINGTABLE

### 5.1.1 DE MODE

| Parameter | Symbol | Spec. | | | Unit |
|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | |
| DCLK Frequency | fclk | 40.8 | 51.2 | 67.2 | MHz |
| Horizontal Display Area | thd | | 1024 | | DCLK |
| HSD Period | th | 1114 | 1344 | 1400 | DCLK |
| HSD Blanking | thb+ thfp | 90 | 320 | 376 | DCLK |
| Vertical Display Area | tvd | | 600 | | $T_H$ |
| VSD Period | tv | 610 | 635 | 800 | $T_H$ |
| VSD Blanking | tvbp+ tvfp | 10 | 35 | 200 | $T_H$ |

### 5.1.2 HV MODE

Horizontal Timing

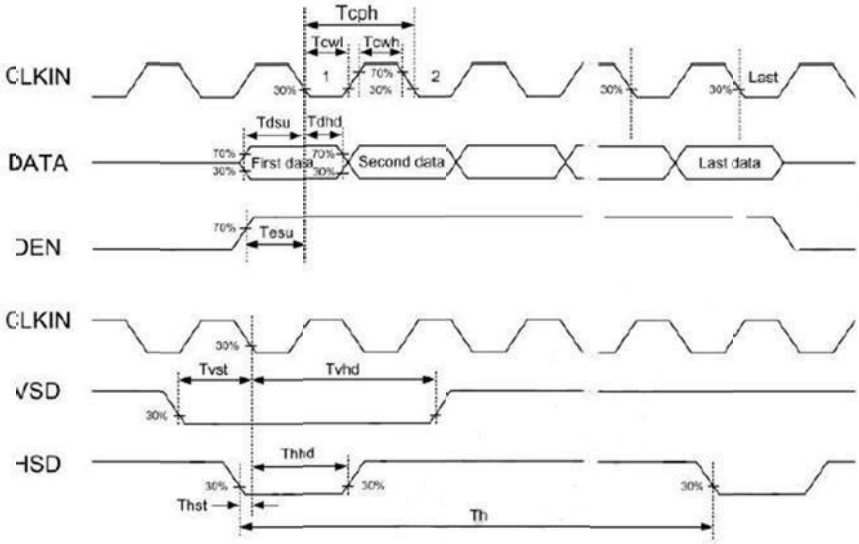
| Parameter | Symbol | Spec. | | | Unit |
|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | |
| DCLK Frequency | fclk | 44.9 | 51.2 | 63 | MHz |
| Horizontal Display Area | thd | | 1024 | | DCLK |
| HS Period | th | 1200 | 1344 | 1400 | DCLK |
| HS Pulse Width | thpw | 1 | - | 140 | DCLK |
| HS Back Porch | thbp | | 160 | | DCLK |
| HS Front Porch | thfp | 16 | 160 | 216 | DCLK |

Vertical Timing

| Parameter | Symbol | Spec. | | | Unit |
|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | |
| Vertical Display Area | tvd | | 600 | | $T_H$ |
| VS Period | tv | 624 | 635 | 750 | $T_H$ |
| VS Pulse Width | tvpw | 1 | - | 20 | $T_H$ |
| S Back Porch | tvbp | | 23 | | $T_H$ |
| VS Front Porch | tvfp | 1 | 12 | 127 | $T_H$ |

## 5.2 Input Clock and Data Timing Diagram

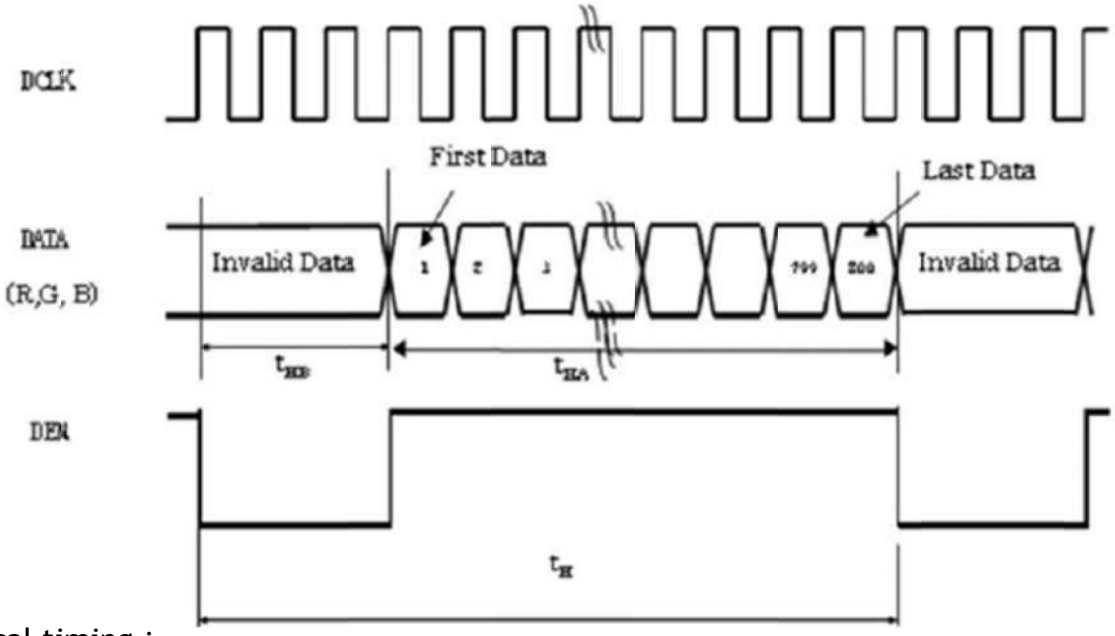
| Parameter | Symbol | Spec. | | | Unit | Condition |
|---|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | | |
| DVDD Power On Slew Rate | TPOR | - | - | 20 | ms | From 0V to 90% DVDD |
| RSTB Pulse Width | TRst | 50 | - | - | us | DCLK=65MHz |
| DCLK Cycle Time | Tcph | 14 | - | - | ns | |
| DCLK Pulse Duty | Tcwh | 40 | 50 | 60 | % | |
| VSD Setup Time | Tvst | 5 | - | - | ns | |
| VSD Hold Time | Tvhd | 5 | - | - | ns | |
| HSD Setup Time | Thst | 5 | - | - | ns | |
| HSD Hold Time | Thhd | 5 | - | - | ns | |
| Data Setup Time | Tdsu | 5 | - | - | ns | D0[7:0],D1[7:0],D2[7:0] to DCLK |
| Data Hold Time | Tdhd | 5 | - | - | ns | D0[7:0],D1[7:0],D2[7:0] to DCLK |
| DEN Setup Time | Tesu | 5 | - | - | ns | |
| DEN Hold Time | Tehd | 5 | - | - | ns | |



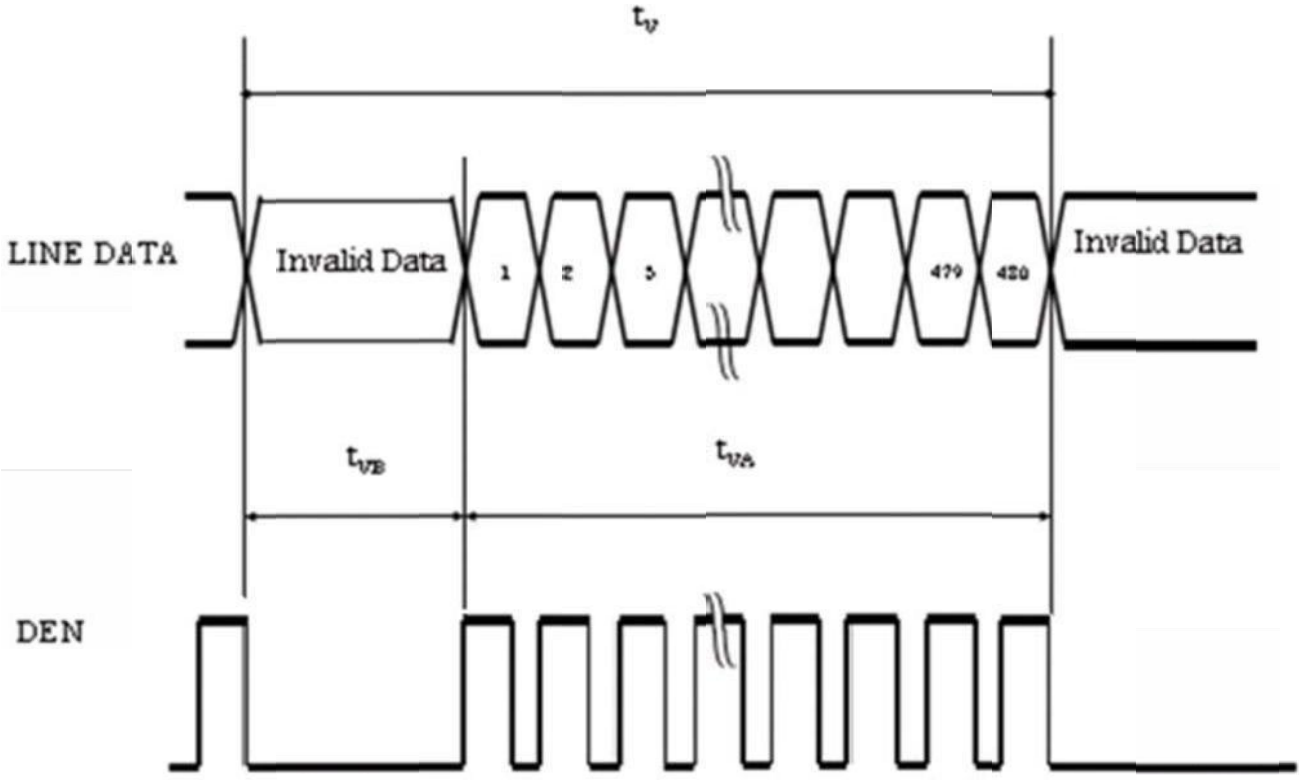
## 6. TIMING SEQUENCE(TIMING CHAR )

### 6.1 DE MODE

Horizontal timing：

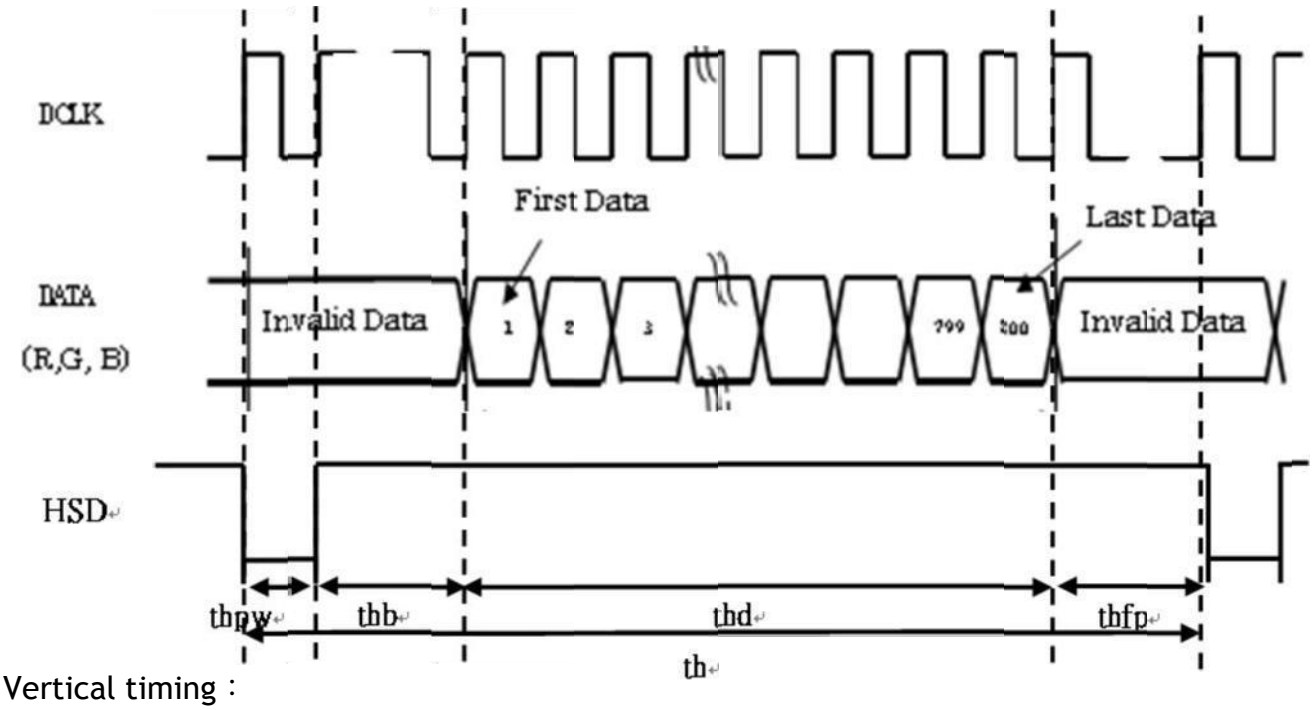



Vertical timing：

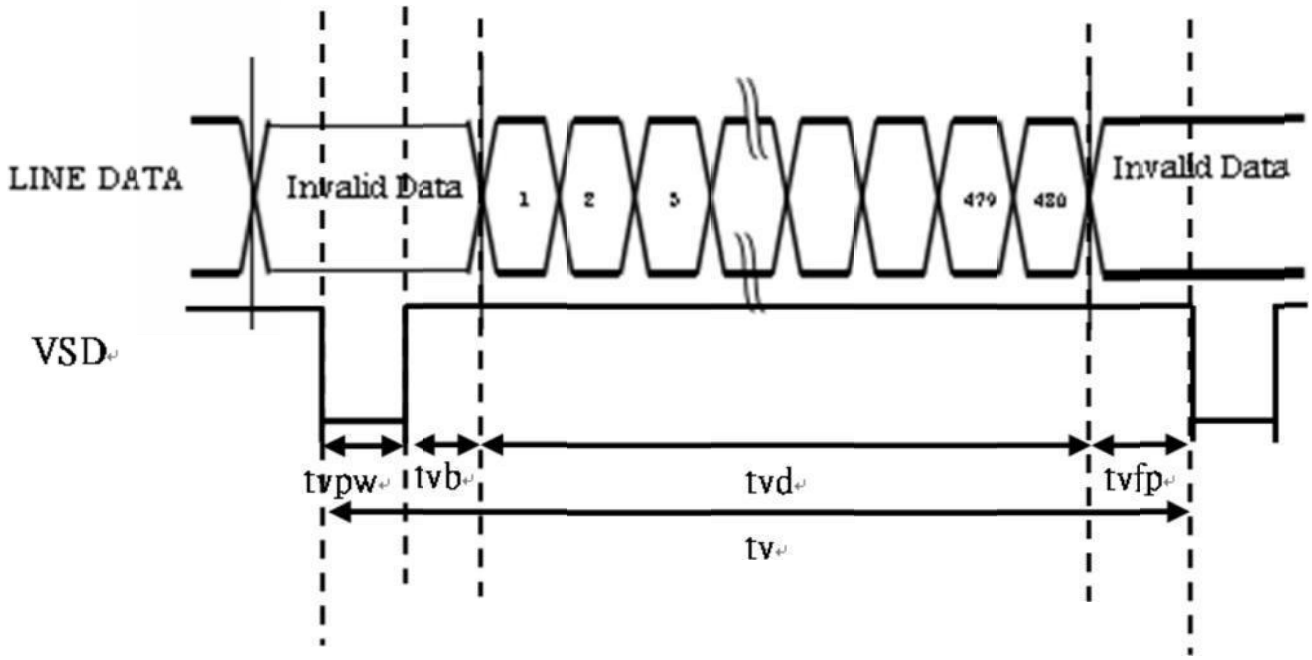


### 6.2 SYNC MODE

Horizontal timing：

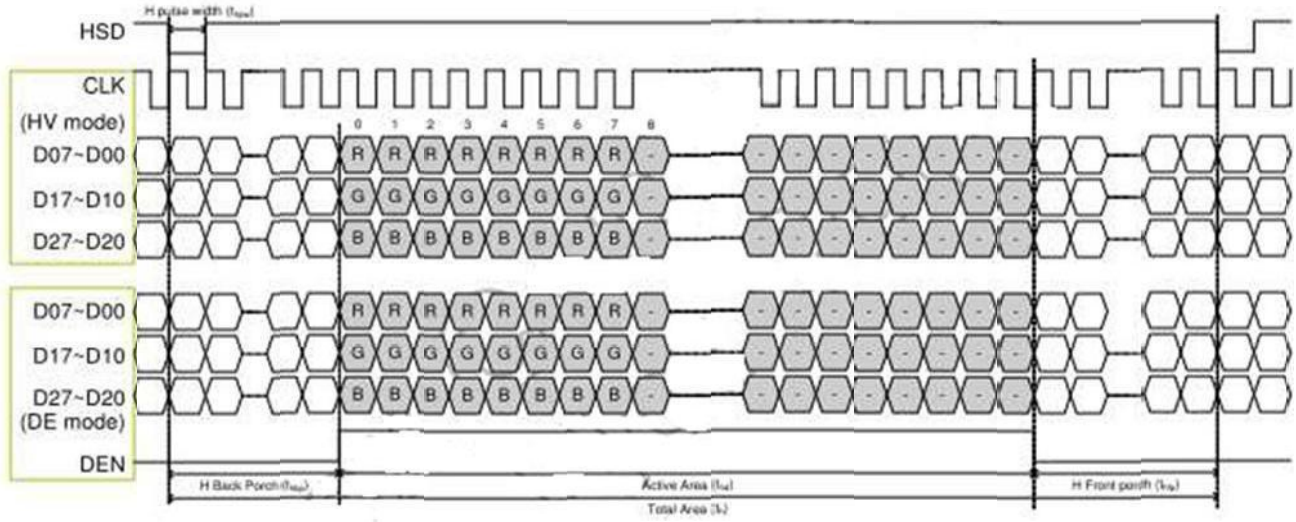



Vertical timing：

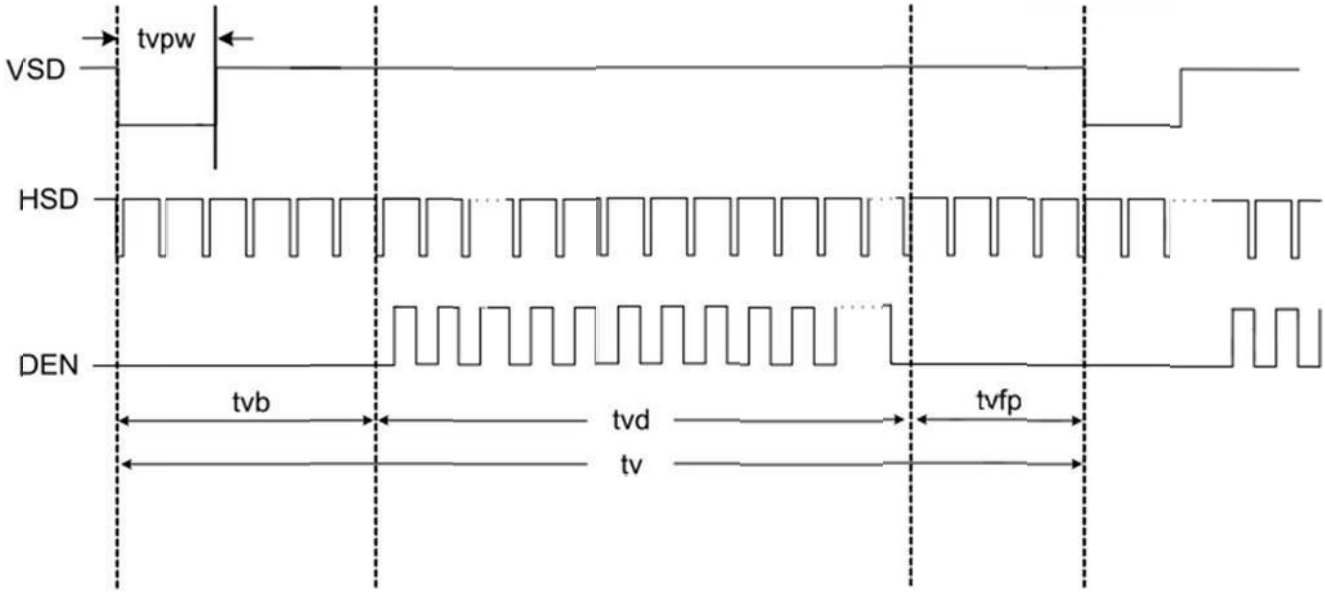


### 6.3 DATA INPUT FORMAT

Horizontal timing：




Vertical timing：



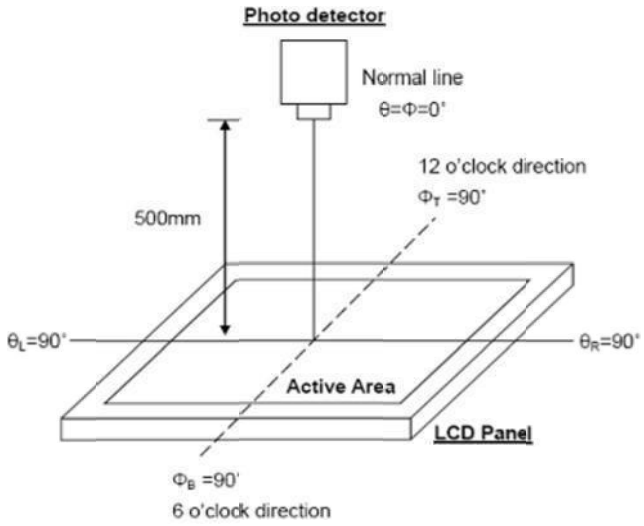
## 7. OPTICAL CHARACTERISTICS

Ta=25℃±2

| Item | | Symbol | Condition | Min. | Typ. | Max. | Unit | Note |
|---|---|---|---|---|---|---|---|---|
| Contrast Ratio | | CR | Θ = 0° | 500 | 600 | - | | Note1 Note4 |
| Luminance | | YL | | 400 | 500 | - | cd/m2 | Note1 Note6 Note7 |
| Luminance Uniformity | | IV-M | | 70 | 75 | | % | |
| Response Time (Rising + Falling) | | $T_{RT}$ | Ta= 25℃ Θ = 0° | - | 25 | 40 | ms | Note1 Note3 |
| Viewing Angle range | Horizontal | ΘL | CR > 10 | | 80 | - | | Note2 |
| | | ΘR | | | 80 | - | | |
| | Vertical | ΘU | | | 80 | - | | |
| | | ΘD | | | 80 | - | | |
| Color Chromaticity | White | x | | 0.230 | 0.280 | 0.330 | | Note1 Note5 Note7 |
| | | y | | 0.260 | 0.310 | 0.360 | | |
| | Red | x | | 0.557 | 0.607 | 0.657 | | |
| | | y | | 0.274 | 0.324 | 0.374 | | |
| | Green | x | | 0.266 | 0.316 | 0.366 | | |
| | | y | | 0.528 | 0.578 | 0.628 | | |
| | Blue | x | | 0.092 | 0.142 | 0.192 | | |
| | | y | | 0.044 | 0.094 | 0.144 | | |
| NTSC | | | | 45 | 50 | | % | |

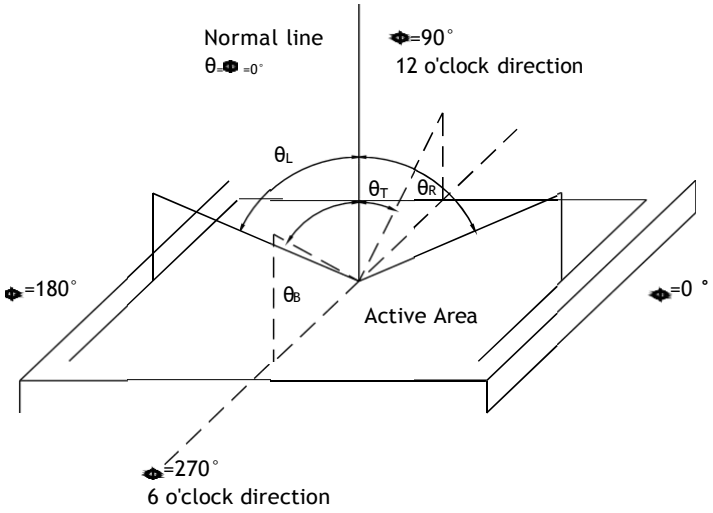
Note1: Definition of optical measurement system

The optical characteristics should be measured in dark room. After 30 minutes operation, the optical properties are measured at the center point of the LCD screen. (Viewing angle is measured by ELDIM-EZ contrast/Height :1.2mm , other items are measured by Photo detector TOPCON BM-7 Field of view: 1° /Height: 500mm.)




Note2: Definition of viewing angle range and measurement system

Viewing angle is measured at the center point of the LCD by CONOSCOPE (ergo-80).




Note3: Definition of Response time

The response time is defined as the LCD optical switching time interval between "White" state and "Black" state. Rise time (TON) is the time between photo detector output intensity changed from 90% to 10%. And fall time (TOFF) is the time between photo detector output intensity changed from 10% to 90%.

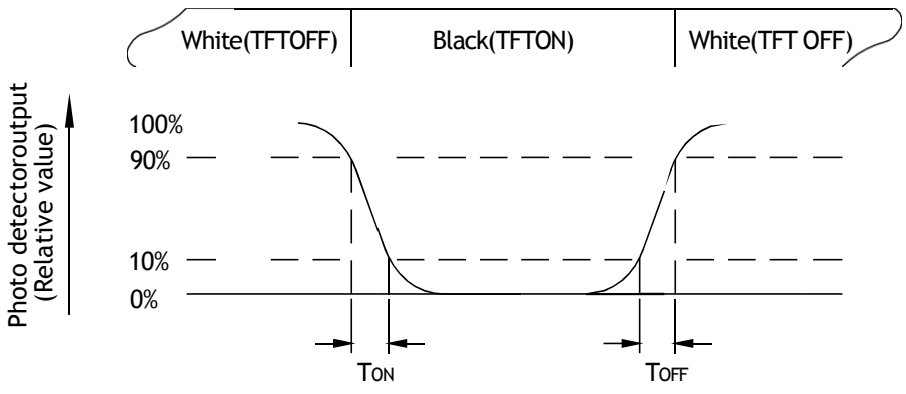



Fig. 6-3 Definition of response time

Note4: Definition of contrast ratio

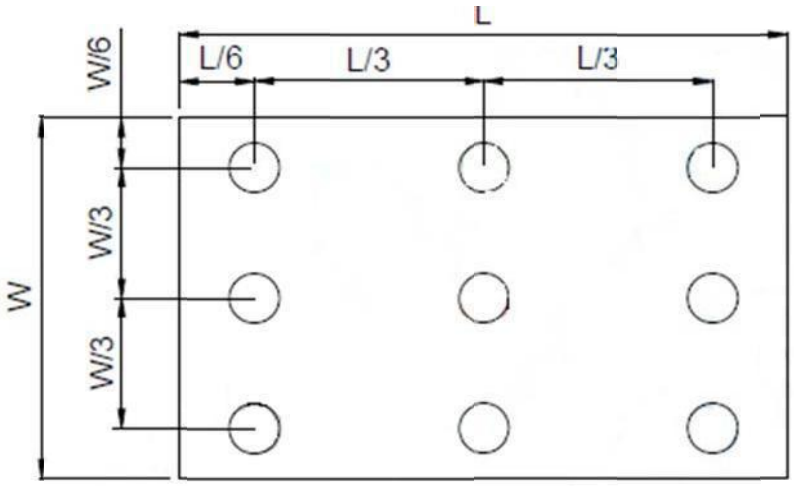
$$\text{Contrast ratio(CR)} = \frac{\text{uminancemeasuredwhen L€Donthe Whitestate}}{\text{Luminancemeasuredwhen L€DontheBlackstate}}$$

"White state ": The state is that the LCD should drive by Vwhite.
"Black state": The state is that the LCD should drive by Vblack.
Vwhite: To be determined Vblack: To be det rmined.

Note5: Definition of color chromaticity (CIE1931)
Color coordinates measured at center point of LCD.

Note6: All input terminals LCD panel must be ground while measuring the center area of the panel. The LED driving condition is IL=140mA.

Note7: Definition of Luminance Uniformity
Active area is divided into 9 measuring areas.Every measuring point is placed at the center of each measuring area.
Luminance Uniformity (U) = Lmin/ Lmax
L----Active area length, W----- Active area width




Bmax: The measured maximum luminance of all measurement position.
Bmin: The measured minimum luminance of all measurement position.

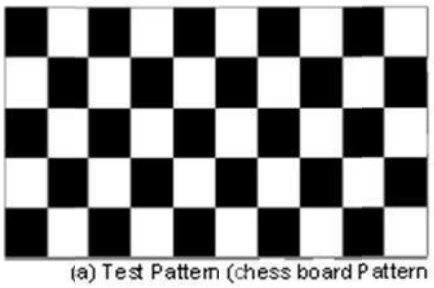
# 8. RELIABILITY TEST ITEMS

## 8.1 TEMPERATURE AND HUMIDITY

| Test Item | Test Condition | Remark |
|---|---|---|
| High Temperature Storage | Ta=55℃; 72hrs | IEC60068-2-1：2007<br>GB2423.2-2008 |
| Low Temperature Storage | Ta=-15℃; 72hrs | IEC60068-2-1：2007<br>GB2423.1-2008 |
| High Temperature Operation | Ta=50℃; 72hrs | IEC60068-2-1：2007<br>GB2423.2-2008 |
| Low Temperature Operation | Ta=-10℃; 72hrs | IEC60068-2-1：2007<br>GB2423.1-2008 |
| High Temperature High Humidity Operation | Ta=45℃，90%RH，72Hrs(no condensation) | IEC60068-2-78：2001<br>GB/T2423.3-2006 |
| Thermal Shock | -15℃(0.5h) ~ 55℃ 0.5h) / 10cycles | Start with cold temperature ，End with high temperature，IEC60068-2-14:1984,GB2423.22-2002 |
| Image Sticking | 25℃ ; 2hrs | Note1 |

Note1:Condition of image sticking test :25ª±2ª

Operation with test pattern sustained for 2hrs,then change to gray pattern immediately.after5 mins,themura must be disappeared completely



(a) Test Pattern (chess board Pattern )



(b) Gray Pattern

## 8.2 VIBRATION&SHOCK

| Test item | Conditions | Remark |
|---|---|---|
| Packing Shock (non-operation) | 980m/s2,6ms, ±x,y,z 3times for direction | IEC60068-2-27：1987<br>GB/T2423.5-1995 |
| acking Vibration (non-operation) | Frequency range:10 HZ~50HZ<br>Stroke:1.0mm,sweep:10 HZ ~50HZ<br>x,y,z 2 hours for each direction | IEC60068-2-32：1990<br>GB/T2423.8-1995 |

## 8.3 ESD

| Test item | Conditions | | Remark |
|---|---|---|---|
| Electro Static Discharge Test (non-operation) | 150pF，330Ω，Contact:±4KV,Air:±8KV | 1 | IEC61000-4-2：2001<br>GB/T17626.2-2006 |
| | 200pF，0Ω，±200V contact test | 2 | |

Note: Measure point :

1. LCD glass and metal bezel
2. IF connector pins

# 9. General Precaution

## 9.1 Safety

1. Do not swallow any liquid crystal, even if there is no proof that liquid crystalispoisonous.
2. If the LCD panel breaks, be careful not to get liquid crystal to touch yourskin.
3. If skin is exposed to liquid crystal, wash the area thoroughly with alcohol orsoap.

## 9.2 Storage Conditions

1. Store the panel or module in a dark place where the temperature is 23±5°C andThehumidity is below 50±20%RH.
2. Store in anti-static electricity container.
3. Store in clean environment, free from dust, active gas, andsolvent.
4. Do not place the module near organics solvents or corrosive gases.
5. Do not crush, shake, or jolt the module.

## 9.3 Handling Precautions

(1) Avoid static electricity which can damage the CMOS LSI.
(2) The polarizing plate of the display is very fragile. So, please handle itverycarefully.
(3) Do not give external shock.
(4) Do not apply excessive force on the surface.
(5) Do not wipe the polarizing plate with a dry cloth, as it may easily scratch theSurface of plate.
(6) Do not use ketonics solvent & Aromatic solvent, use with a soft cloth soaked witha cleaning naphtha solvent.
(7) Do not operate it above the absolute maximum rating.
(8) Do not remove the panel or frame from the module.
(9) When the module is assembled, it should be attached to the system firmly,Be careful not to twist and bend the module.
(10) Wipe off water droplets or oil immediately. If you leave the droplets for a long time,staining and discoloration may occur.
(11) If the liquid crystal material leaks from the panel,it should be kept away from theeyes or mouthIn case of contact with hands,legs or clothes, it must be washedawaythoroughly with soap.

## 9.4 Warranty

(1) The period is within twelve months since the date of shipping out under normal using and storage conditions.
(2) Do not repaired or modified the LCM. It may cause function to lose efficacy, BOHU does not warrant the LCM.
(3) All process and material comply ROHS.

## 10. PACKAGE DRAWING

